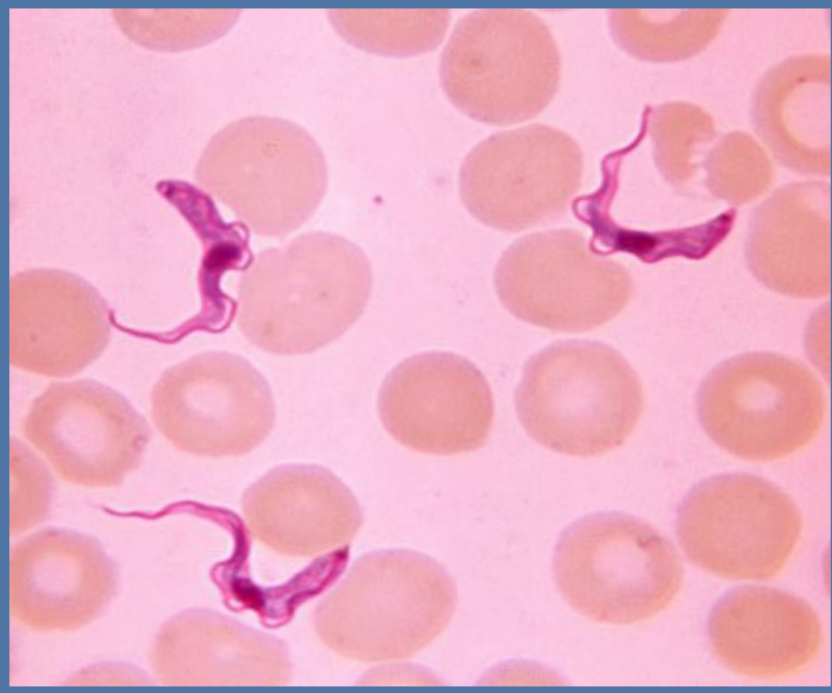




LIGHTWEIGHT CNNs FOR TRYPANOSOME DETECTION

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Scan for Demo here

Introduction

Trypanosoma parasites cause serious diseases like African Sleeping Sickness and Chagas Disease, which are nearly always fatal without treatment. Diagnosis depends on trained microscopists manually examining blood smears – a slow, error-prone process that requires specialized expertise unavailable in the rural, resource-limited regions where these diseases are most prevalent. Existing AI-based detection systems have demonstrated strong accuracy but rely on GPU hardware and complex pipelines that aren't deployable in field clinics. To address this gap, we trained a lightweight YOLOv7-tiny object detection model capable of running on any standard laptop without a GPU, making automated screening accessible to the settings where it's most needed.

Novelty

Most published trypanosome detection systems, including the high-accuracy hybrid model from Kittichai et al. (2023), rely on heavy computational resources and are confined to laboratory environments. Our approach uses YOLOv7-tiny – a compact architecture with only 6 million parameters, roughly 6× smaller than the full YOLOv7 – fine-tuned via transfer learning from COCO pretrained weights. The trained model is exported to ONNX format and runs on standard laptop CPUs using only three Python packages (onnxruntime, opencv-python, numpy). The resulting 12 MB model is small enough to be distributed via email or USB drive, making it practical for deployment in clinics that lack specialized hardware, GPU access, or stable internet connections.

Training Pipeline

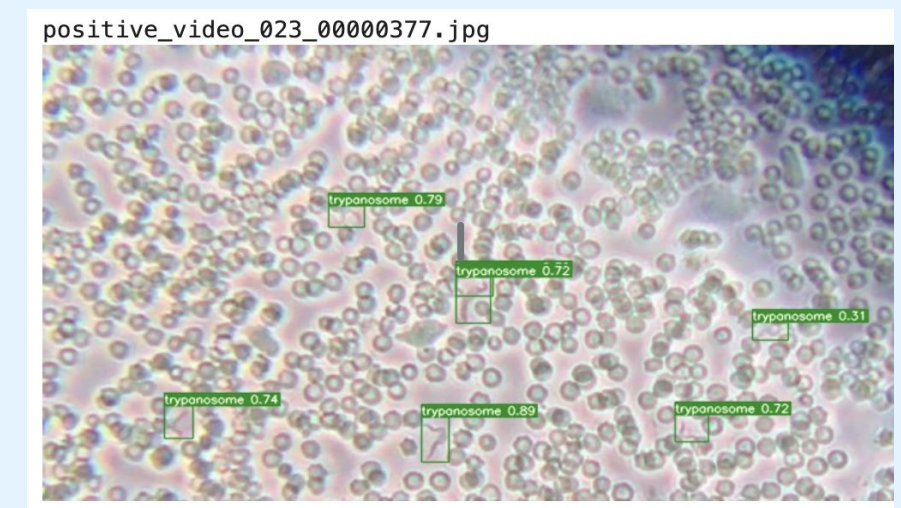
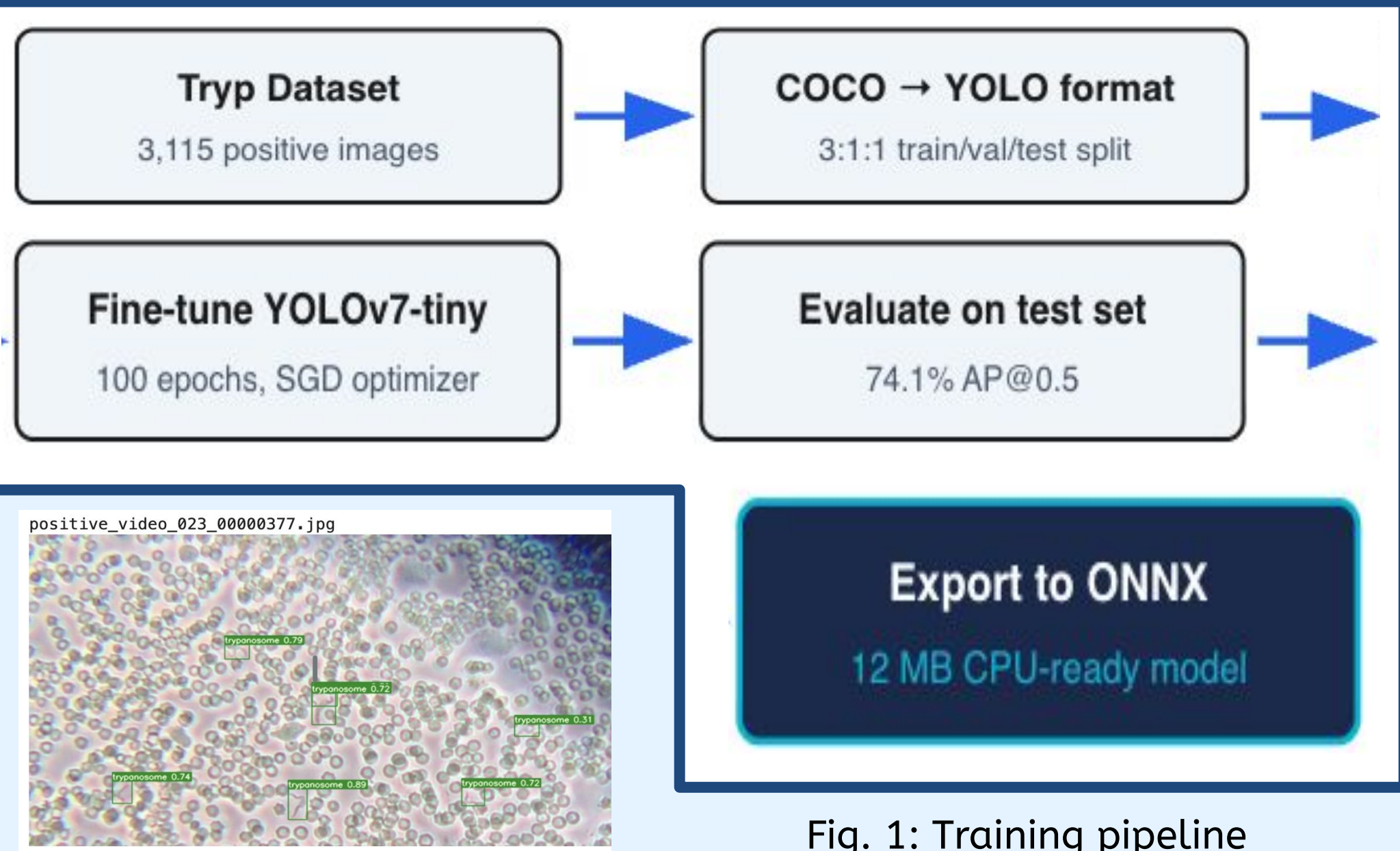


Fig. 2: Sample Annotated Image

Fig. 1: Training pipeline

Detection Pipeline

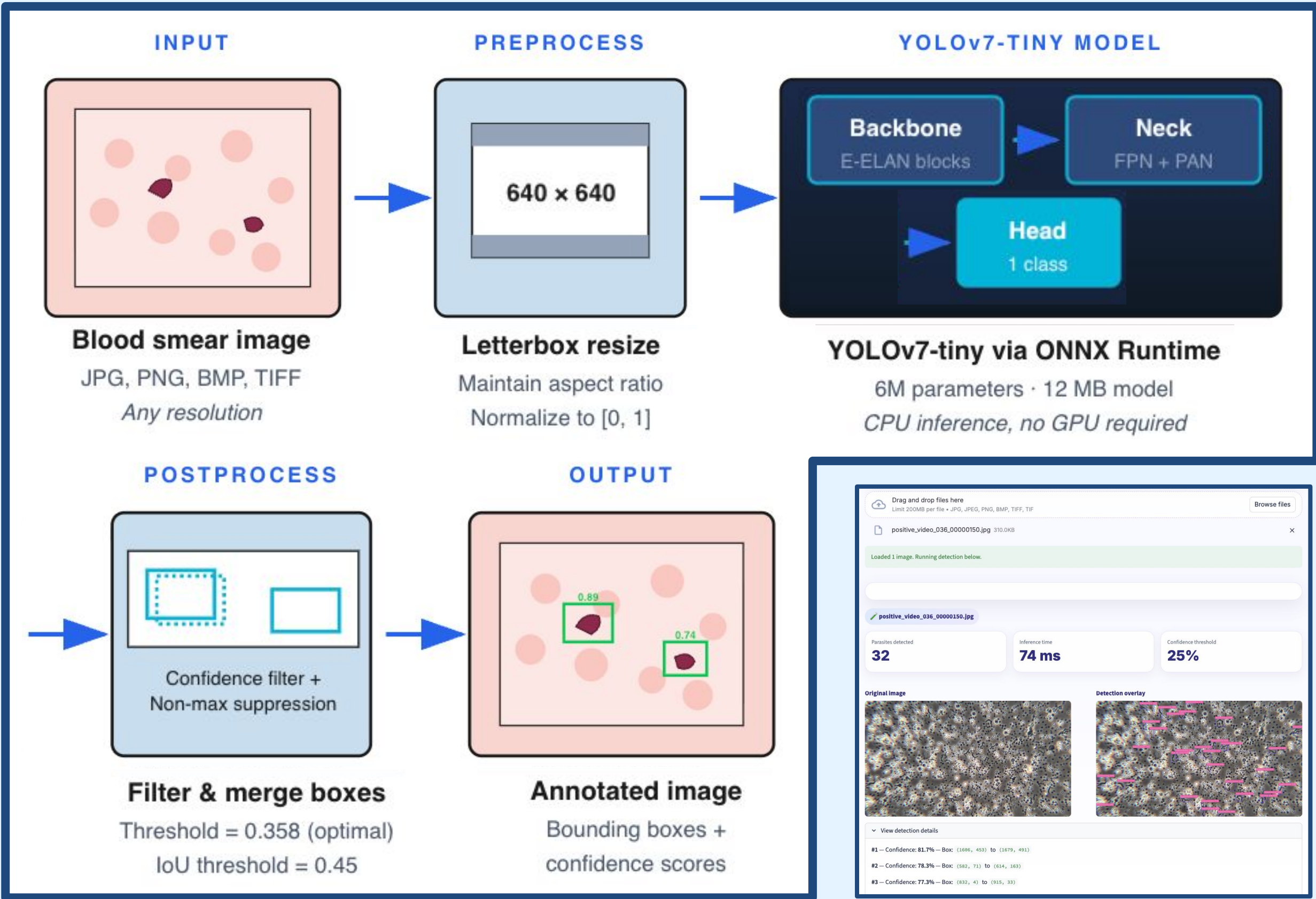


Fig. 3: Inference pipeline

Fig. 4: Web interface snippet

Methodology

We trained on the Tryp dataset (Anzaku et al., 2023), which contains over 3,000 microscopy images of unstained thick blood smears with bounding box annotations for the Trypanosoma brucei parasite. Fig 2. shows a sample annotated image.

After converting the COCO-format annotations into YOLO format, we split the dataset 3:1:1 into training, validation, and test sets (see Fig. 1).

We used YOLOv7-tiny, a single-stage object detector with three main components: a backbone of efficient layer aggregation blocks (E-ELAN) that extract multi-scale features, a neck combining a Feature Pyramid Network and Path Aggregation Network that fuses features across scales, and a detection head that produces bounding box coordinates, objectness scores, and class probabilities at each cell.

Training was performed on a free Google Colab T4 GPU for 100 epochs with a batch size of 16 at 640×640 resolution, using SGD with a learning rate of 0.01 and momentum of 0.937, starting from COCO pretrained weights. After training, the best checkpoint was exported to ONNX format using YOLOv7's built-in export pipeline (see Fig. 1).

A standalone CPU inference script and a Streamlit web interface (Fig. 4) were also developed to demonstrate end-to-end deployment (see Fig. 3).

Detection Pipeline

The trained model achieved 74.1% AP@0.5, 74.9% precision, 70.2% recall, and 34.4% AP@0.5:0.95 on the held-out test set of 630 images. The precision-recall curve in Fig. 5 shows the tradeoff between precision and recall, and the area under this curve gives us mAP at 0.5 IoU, which for our model is 74.1%. F1 analysis identified an optimal confidence threshold of 0.358, with peak F1 of 0.72 (see Fig. 6). These results are competitive with most published baselines on this dataset, despite using a model with far fewer parameters than the full YOLOv7. The exported ONNX model is only 12 MB and runs efficiently on a laptop CPU, demonstrating that competitive trypanosome detection is achievable on accessible consumer hardware.

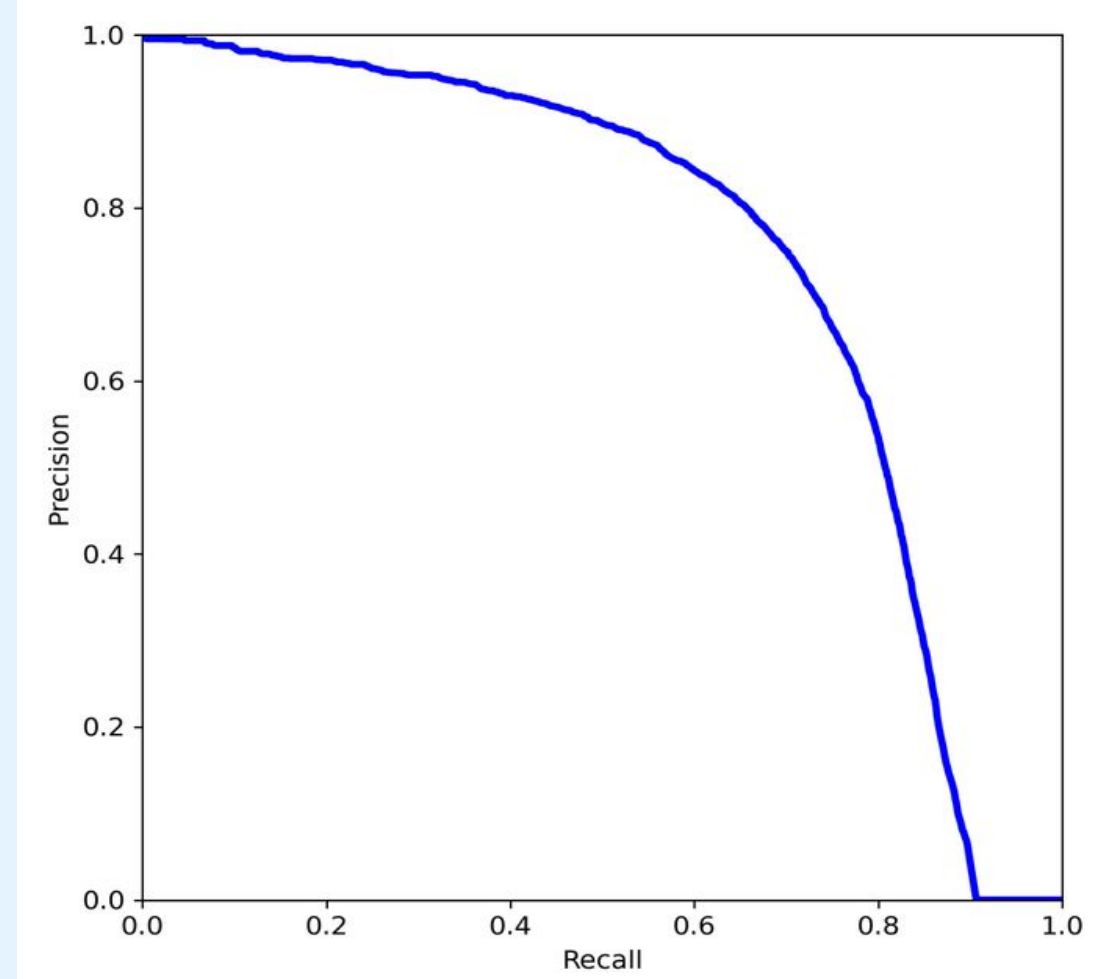


Fig. 5: Recall v Precision Graph

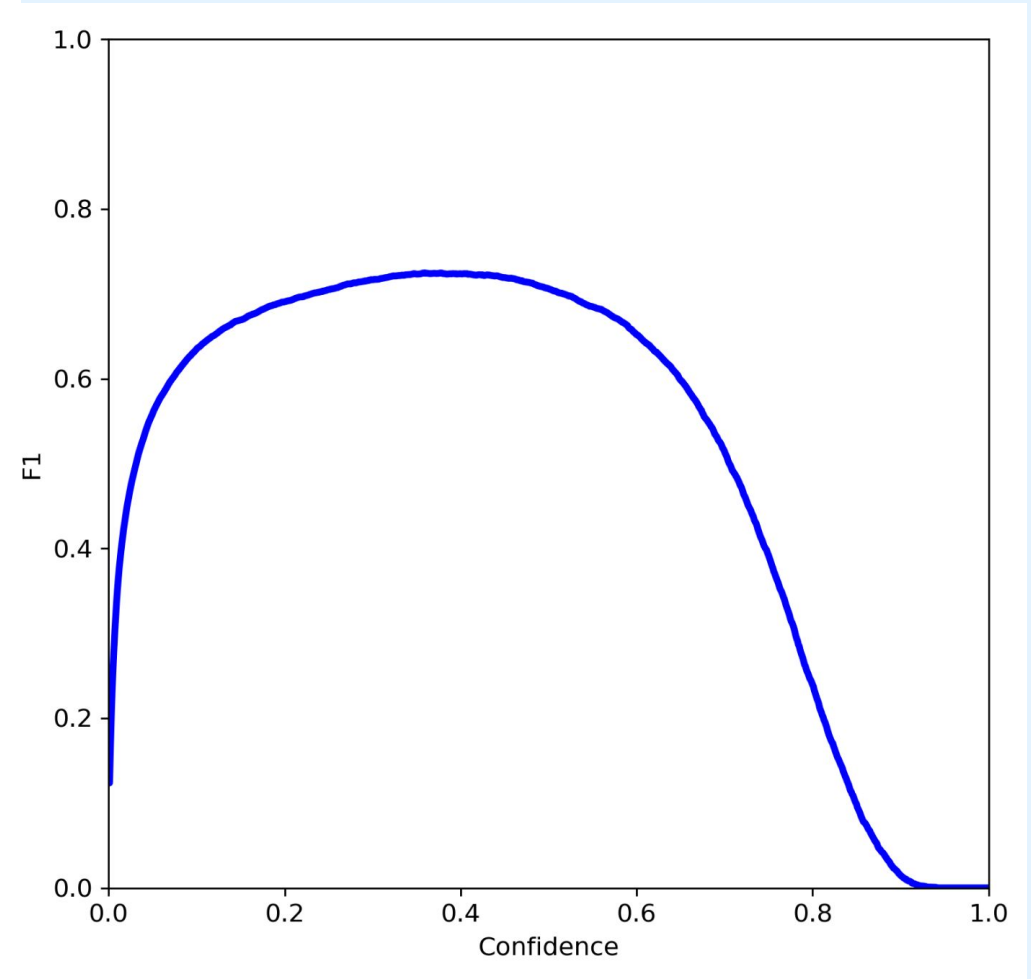


Fig. 6: Confidence v F1 Graph

Impact & Future Work

By making accurate trypanosome detection possible on standard laptops, this work removes a major barrier to deployment in rural sub-Saharan African clinics where HAT screening is most needed. Even at 70% recall, automated screening represents a significant improvement over zero coverage in clinics without trained microscopists, serving as a first-line tool that flags potentially positive samples for human confirmation. Future work includes expanding to multi-species classification (T. cruzi, T. evansi), packaging the system as a mobile application, and benchmarking newer lightweight architectures such as YOLOv8-nano and YOLO11.

References

Anzaku, E. T., Mohammed, M. A., Ozbulak, U., Won, J., Hong, H., Krishnamoorthy, J., Van Hoecke, S., Magez, S., Van Messem, A., & De Neve, W. (2023). Tryp: A dataset of microscopy images of unstained thick blood smears for trypanosome detection. Scientific Data, 10, 716. <https://doi.org/10.1038/s41597-023-02608-y>

Kittichai, V., Kaewthamasorn, M., Thanee, S., Sasisaowapak, T., Naing, K. M., Jomtarak, R., Tongloy, T., Chuwongin, S., & Boonsang, S. (2023). Superior auto-identification of trypanosome parasites by using a hybrid deep-learning model. Journal of Visualized Experiments, (200) <https://doi.org/10.3791/65557>